

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 07)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

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2

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3

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0 1 . 2

Explain how an oil spill at sea can kill seabirds.

[3 marks]

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0 1 . 3

Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

[2 marks]

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0 1 . 4

Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

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10

0 2 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

0 2 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

0 2 . 3 Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

0 2 . 4 Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

10

03

A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha ⁻¹	Carbon in soil dead organic matter / t ha ⁻¹
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

03

. 1

Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer

t/ha

03

. 2

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

03

. 3

Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1

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2

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5

0 4 . 1

Explain how energy absorbed in tropical regions is distributed to higher latitudes such as the UK.

[4 marks]

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0 4 . 2

Explain why the UK has milder winters than Moscow or Churchill, Canada, which are at similar latitudes.

[2 marks]

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0 4 . 3

Explain why the atmosphere would not retain liquid water if atmospheric pressure were much lower.

[2 marks]

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0 4 . 4

Give **two** industrially important gases extracted from the atmosphere.**[2 marks]**

1

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2

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10

0 5

Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

Table 1

	Unmetered households (fixed charge)	Metered households
Mean water use / dm ³ per person per day	150	125
Percentage with low-flush toilets	35	58
Percentage using a garden hosepipe weekly in summer	44	20

0 5 . 1

Calculate the percentage reduction in water use per person in metered households.

[1 mark]

Answer

%

0 5 . 2

Explain why metering reduces water use.

[2 marks]

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0 5 . 3

Describe **three** ways in which domestic water use can be reduced other than metering.

[3 marks]

1

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2

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3

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0 5 . 4

Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

[2 marks]

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0 5 . 5

Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

[2 marks]

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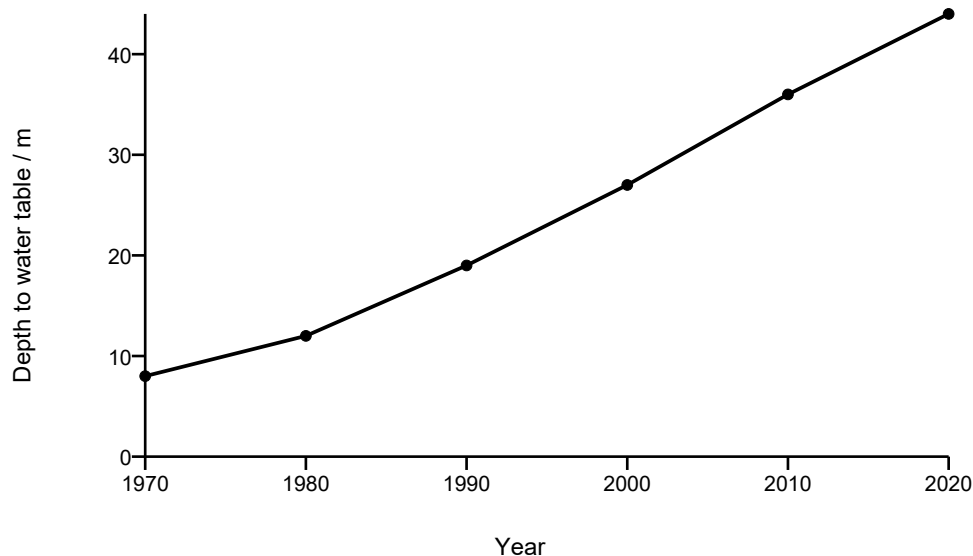
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0 6

Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1**0 6****. 1**

Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]**Answer**

m/yr

0 6**. 2**

Explain why the water table is falling.

[2 marks]

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0 6**. 3**

Explain the features a rock must have to form an aquifer.

[3 marks]

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0 6 . 4 Describe **four** consequences of over-exploiting this coastal aquifer.

[4 marks]

1

2

3

4

10

0 7 Table 1 shows the per capita energy consumption and GDP per capita of five countries.

Table 1

Country	GDP per capita / US\$ thousand	Energy use per capita / GJ per year	Main energy resource
A	62	290	Oil, gas
B	44	125	Gas, nuclear, renewables
C	10	95	Coal
D	2	25	Biomass (wood), oil
E	55	220	Hydroelectricity, oil

0 7 . 1 Describe the relationship between GDP per capita and energy use per capita shown in Table 1.

[2 marks]

07 . 2

Explain why affluence increases per capita energy consumption.

[3 marks]

07 . 3

Suggest **two** reasons why country B uses much less energy per capita than country A despite a similar level of affluence.

[2 marks]

1

2

07 . 4

Explain why the per capita energy consumption of the UK has fallen in recent years even though consumption of goods has risen.

[2 marks]

07 . 5

Give **one** way energy is used in each of the following: water supplies; agriculture.

[1 mark]

10

08

Table 1 shows measurements taken from a stream above and below the point where drainage water from an old metal-mine spoil heap enters it.

Table 1

Site	pH	Dissolved zinc / mg dm ⁻³	Dissolved oxygen / % saturation	Number of invertebrate taxa	Mayfly larvae present?
Upstream	7.1	0.02	96	18	Yes
50 m downstream	3.4	6.5	94	3	No
2 km downstream	5.9	1.1	95	9	No

08

1

Explain how the acidic drainage water is produced in the spoil heap.

[3 marks]

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08

2

Use Table 1 to explain why the number of invertebrate taxa falls so sharply 50 m downstream, even though dissolved oxygen is unchanged.

[3 marks]

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08

3

Suggest why the pH and zinc concentration have partly recovered 2 km downstream.

[2 marks]

08 . 4

Describe how acid mine drainage can be controlled.

[2 marks]

10

09 . 1

Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

Waste category	Example / source	Storage or disposal method
High-level waste		
Intermediate-level waste		
Low-level solid waste		

09 . 2

Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

09 . 3

Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

10

10 . 1

Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

Do not write outside the box

Extra space

10 . 2

State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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END OF QUESTIONS